

# A dot product asks one question: **are these two arrows pointing the same way?**

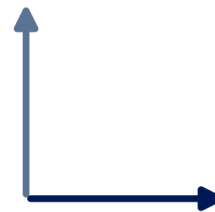
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**DOT PRODUCT = HOW MUCH DO THEY AGREE?**



same way

**+1**



right angles

**0**



opposite

**-1**

- 1 Two vectors pointing the **same way** give a big positive number.
- 2 **At right angles** gives zero — no agreement at all.
- 3 **Opposite ways** gives a negative number. That is the whole intuition.

**REMEMBER THIS**

**Every similarity score in AI — search, recommendations, embeddings — is this one question.**